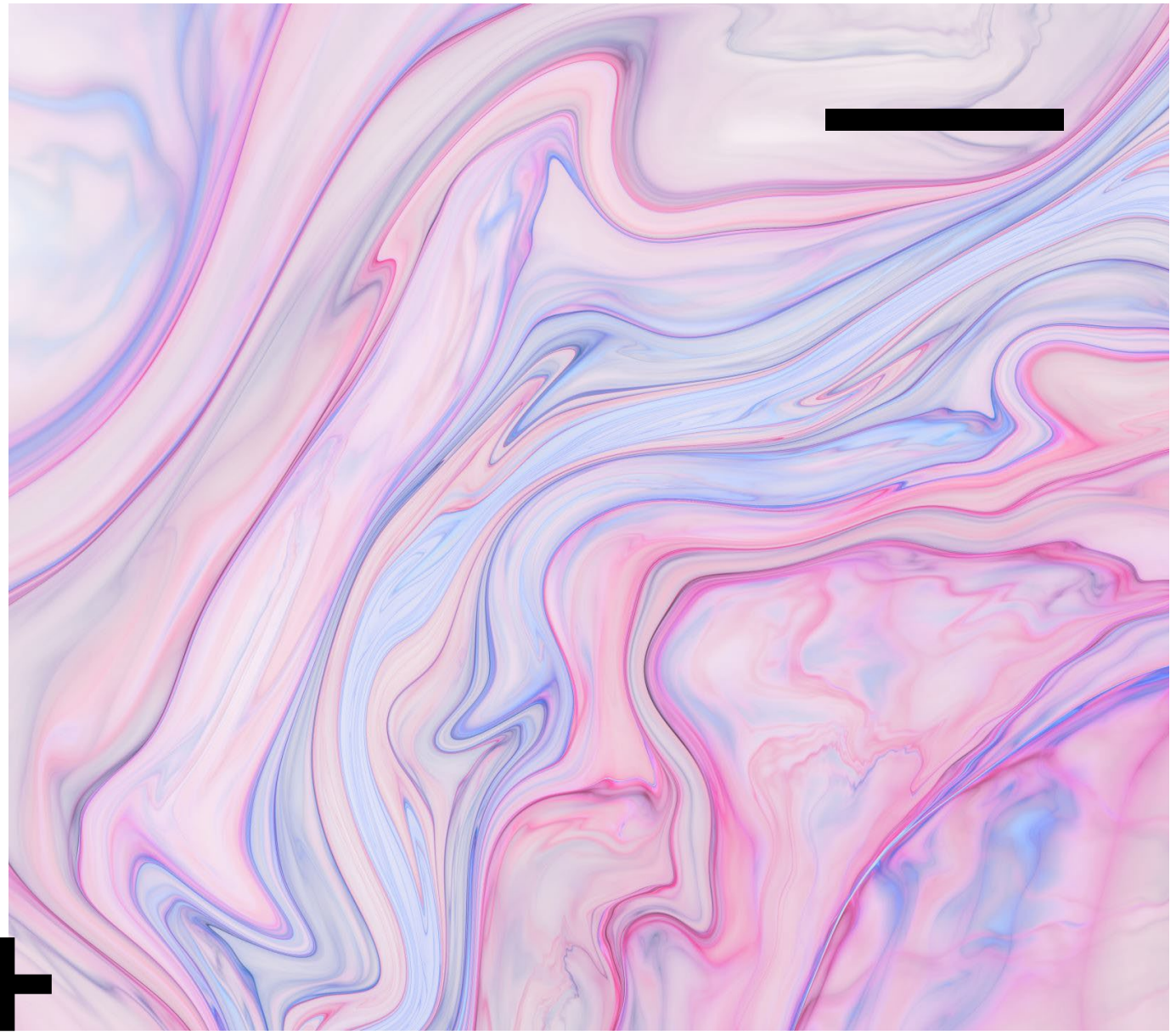
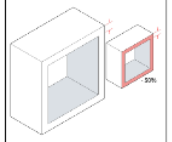
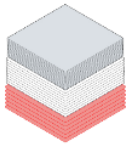
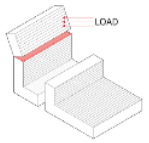
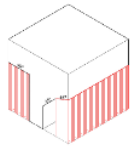
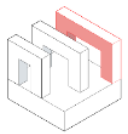
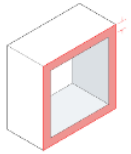
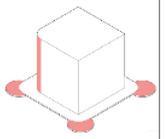
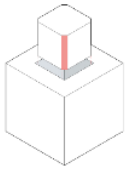
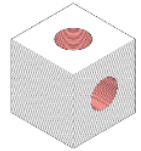
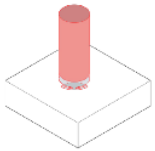
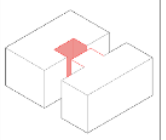
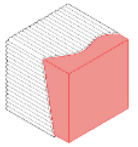
NBD1100 Built Environment Communication and Skills 3D Printing Modelling Guide

3D Printing Presentation 2

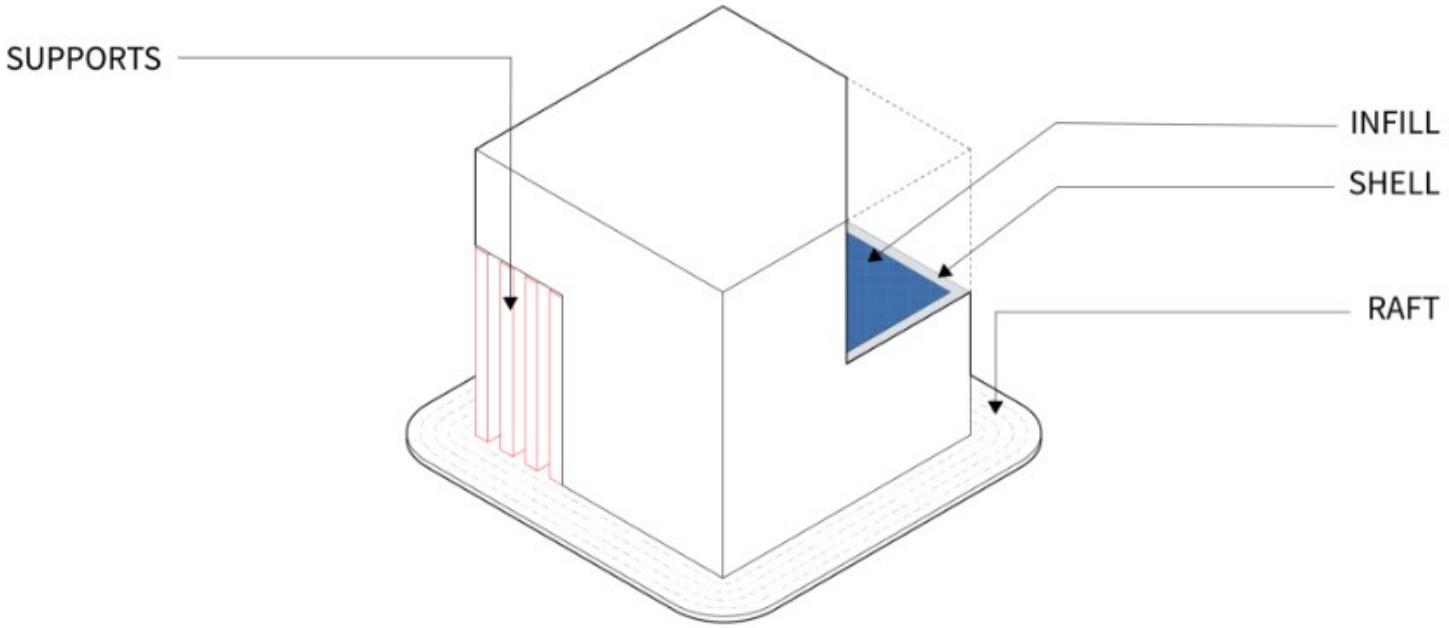
by Shiran Geng



Summary from Presentation 1

	SCALE	LAYER RESOLUTION	LAYER STRENGTH/ ORIENTATION	SUPPORTS AND OVERHANGS	HORIZONTAL BRIDGES	WALL THICKNESS	LARGE SURFACE AREA	CORNERS	EMBOSSED DETAILS	HOLE ORIENTATION	INTERLOCKING/ MOVING PARTS	TOLERANCE	POST PROCESSING/ FINISHES
DIAGRAM													
CONCEPTUAL MODEL	PART / ISOLATED SECTION	RESOLUTION IS LESS IMPORTANT WHEN PRINTING CONCEPT MODELS	CONSIDER IF THIS STRENGTH OF THE MODEL IS IMPORTANT IN YOUR OUTCOME	ENSURE SUCESS OF THE PRINT WITH SUPPORTS	MAXIMISE THE USE OF BRIDGES TO OPTIMISE PRINT TIME	2MM MINIMUM THICKNESS WILL ENSURE SUCCESS OF ALL PRINTS	QUESTION THE NEED TO PRINT A LARGE/WHOLE MODELS AT THIS STAGE	IMPORTANCE OF CORNERS IS SUBJECT TO DESIGN INTENT AND OUTCOME	DETAILS CAN BE OMITTED IF UNIMPORTANT TO FINAL OUTCOME	ORIENTATION AND DESIGN CAN REFINE MODEL OUTCOME	NOT NECESSARY UNLESS REQUIRED IN FINAL OUTCOME	NOT NECESSARY UNLESS REQUIRED IN FINAL OUTCOME	UNLIKELY TO REQUIRE POST PROCESSING
STUDY MODEL	WHOLE / PART	RESOLUTION DEPENDENT ON INTENDED QUALITY OF MODEL	STRENGTH IN X-Y DIRECTION CAN BE EXPLORED DURING THE INITIAL DESIGN STAGES	IDENTIFY TIME CONSUMED WHEN PRINTING AND REMOVING SUPPORTS	QUALITY OF BRIDGES IS INSIGNIFICANT UNLESS IT IS PART OF THE DESIGN INTENT	2MM MINIMUM, HOWEVER FINER DETAILS CAN BE EXPLORED AT THIS STAGE	ENSURE YOU EXPLORE THE SUCCESS OF LARGER PRINTS BEFORE FINALISING YOUR PRINT OUTCOME	IMPORTANCE OF CORNERS IS SUBJECT TO EXPLORATION OF MODEL INTENT	TESTING OF DETAILS IS IMPORTANT TO DETERMINE IF NECESSARY	QUALITY OF HOLES CAN BE EXAMINED WHEN PRINTING STUDY MODELS	INSPECT THE NEED FOR MOVING OR INTERLOCKING PARTS	ENSURE ALL TOLERANCES ARE CHECKED AS THIS WILL AFFECT FINAL OUTCOME	POST PROCESSING CAN BE TESTED ON PART OF THE MODEL TO STUDY OUTCOMES
PRESENTATION MODEL	WHOLE / PART	GOOD RESOLUTION CAN BE ACHIEVED BY LOWERING LAYER HEIGHT	IMPORTANCE OF STRENGHT IS DEPENDENT ON DESIGN INTENT	MINIMISING SUPPORTS WILL MINIMISE SCARRING OF FINAL MODEL	USE HORIZONTAL BRIDGES WISELY	UNLESS ADVISED, ENSURE GEOMETRY IS 2MM MINIMUM	EXPLORE OTHER OPTIONS BEFORE PRINTING LARGE AREAS AS IT PROVES A HIGHER RISK	CORNERS CAN BE IMPROVED IN FINAL OUTCOMES WHEN TESTED	FINAL DETAILS SHOULD BE EXPLORED BEFORE PRINTING A PRESENTATION MODEL	ORIENTATING HOLES IN THE Z DIRECTION CAN IMPROVE SMOOTHNESS OF MODEL	UNLESS INTENEDED, INTERLOCKING PARTS SHOULD BE AVOIDED IN FINAL OUTCOME TO PREVENT FAULTS	ENSURE TOLERANCE OF MODEL HAS BEEN CHECKED BEFORE DECIDING TO CONTNUE	FINAL MODELS CAN BE IMPROVED WITH FINISHED TO ELEVATE OUTCOMES
FUNCTIONAL MODELS	WHOLE / PART	ENSURE THE RESOLUTION OPTIMAL FOR MODEL TO BE FUNCTIONAL	DOUBLE CHECK PRINTING ORIENTATION AND GEOMTRY IS OPTIMAL	ALWAYS PRINT SUPPORTS TO AID MODEL SUCCESS	ENSURE THAT SUPPORTS UNDER BRIDGES IS TURNED ON FOR A BETTER OUTCOME	ENSURE MODELS SUCCESS BY SUPPORTING STRENGHT IN GEOMETRY THICKNESS	WHEN PRINTING LARGE SURFACES, MAKE SURE HELPER DISKS ARE ADDED TO SUPPORT THE PRINT OUTCOME	ENSURE CORNERS ARE DONE WITH CARE	UNLESS INTENTED, DETAILS CAN BE MINIMISED TO BETTER FOCUS ON FUNTIONALITY OF THE MODEL	HOLE ORIENTATION CAN IMPROVE TOLERANCES ON THE MODEL	ENSURE THAT TESTING HAS BEEN CARRIED OUT	TOLERANCES SHOULD BE CHECKED ON ALL PARTS TO PREVENT DEFECTS WHEN PRINTING	MODELS CAN ALSO BE STRENGTHEN WITH POST PROCESSING AND FINISHES IF REQUIRED

+



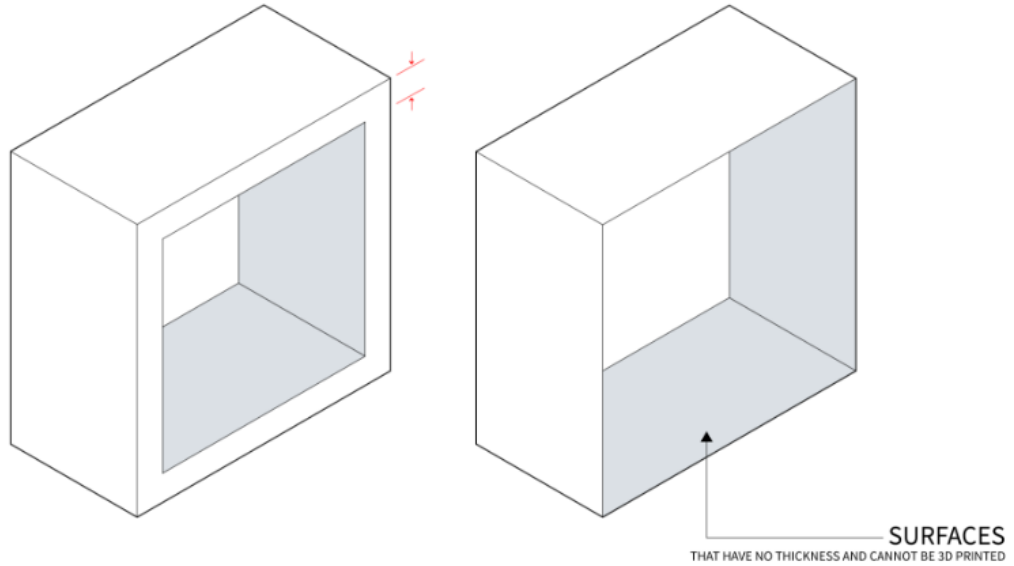
key elements of a 3D print

As with all tools, it is important to understand how the functions of a 3D printer can impact the possibilities of your printed geometry.

The diagram below shows the key elements of a 3D print. For the most optimal result, the following capabilities and limitations should be thoroughly considered when designing a model.

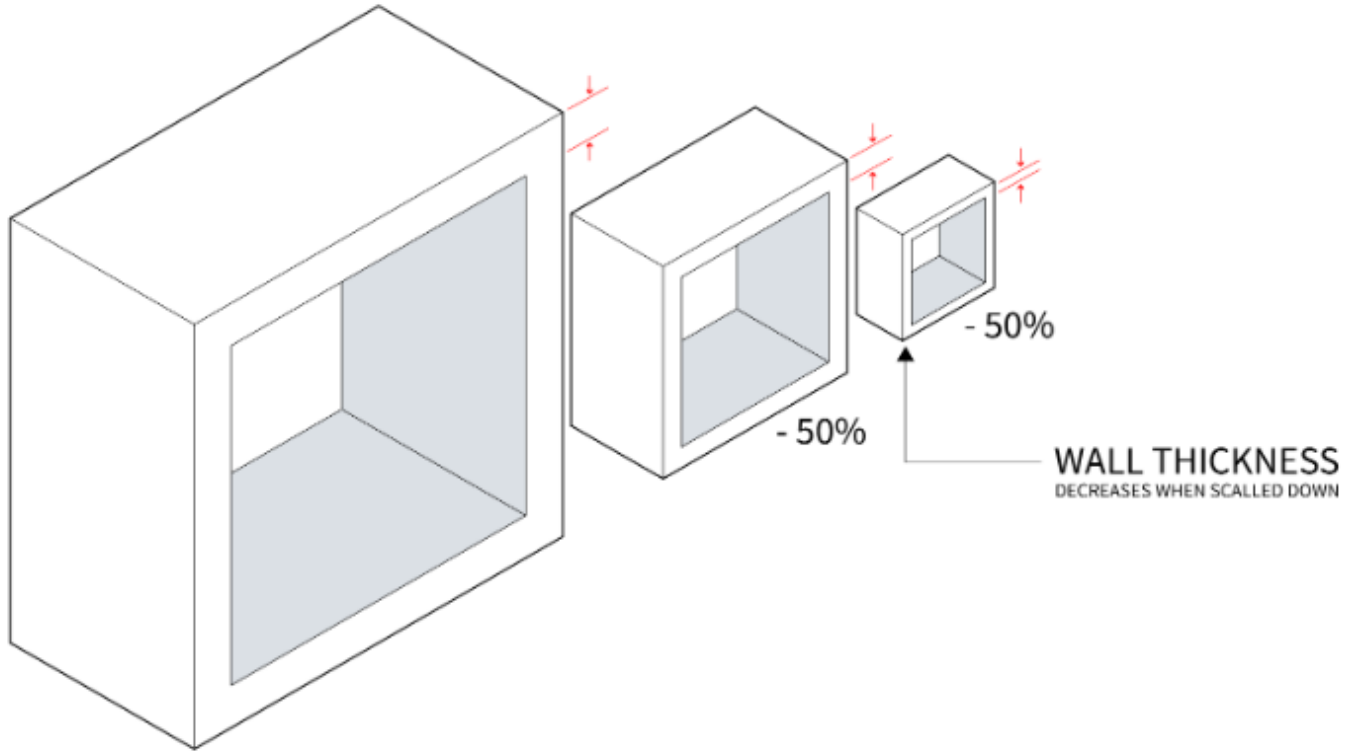
1. Volumes vs Surfaces

- 3D printing generally requires solid geometry with thickness to print. To a computer, a surface is actually an infinitely thin object and will not be understood as 3D printing geometry.
- In Rhino, there are some commands to get you from surfaces to closed polysurfaces, including the commands [ExtrudeSrf], [OffsetSrf] and [OffsetMesh] for Meshes.



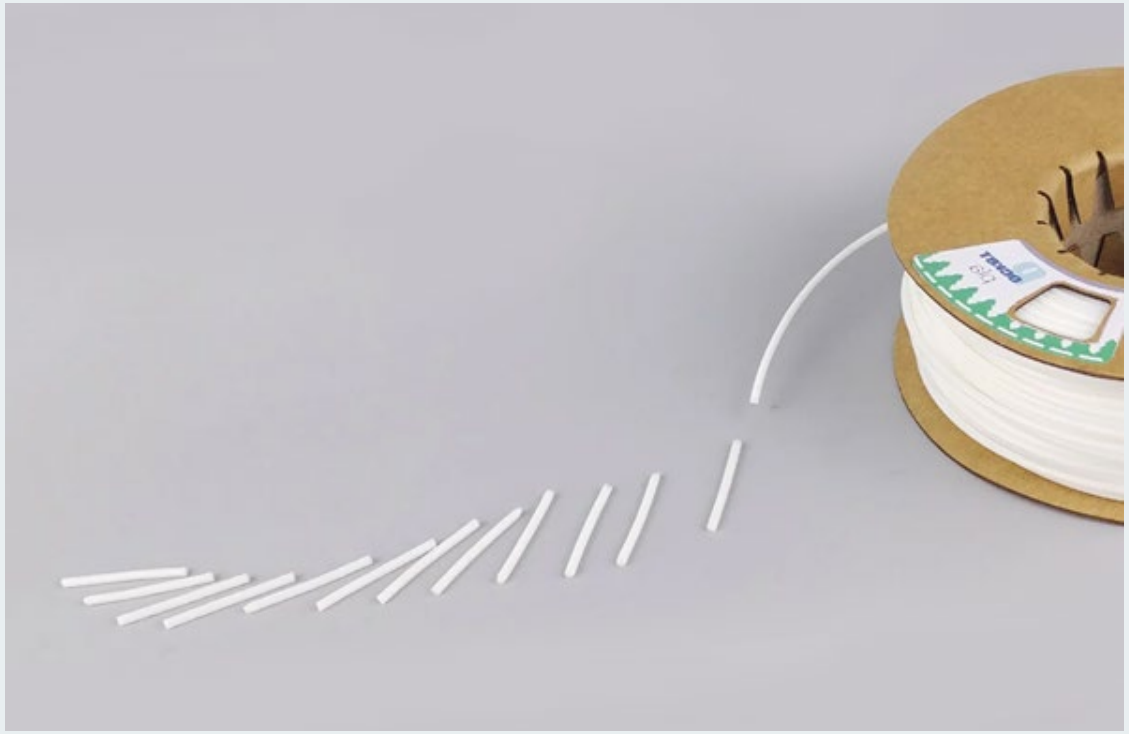
2. Scaling and Detail

- When scaling models, it is important to bear in mind the minimum thicknesses and detail resolution for 3D printing. For that reason, it is recommended that you design your model within the scale that you have decided instead of simply scaling down a full scale sized model.
- If you have an existing model, you can begin by scaling it down but you will need to adjust the model to these constraints.



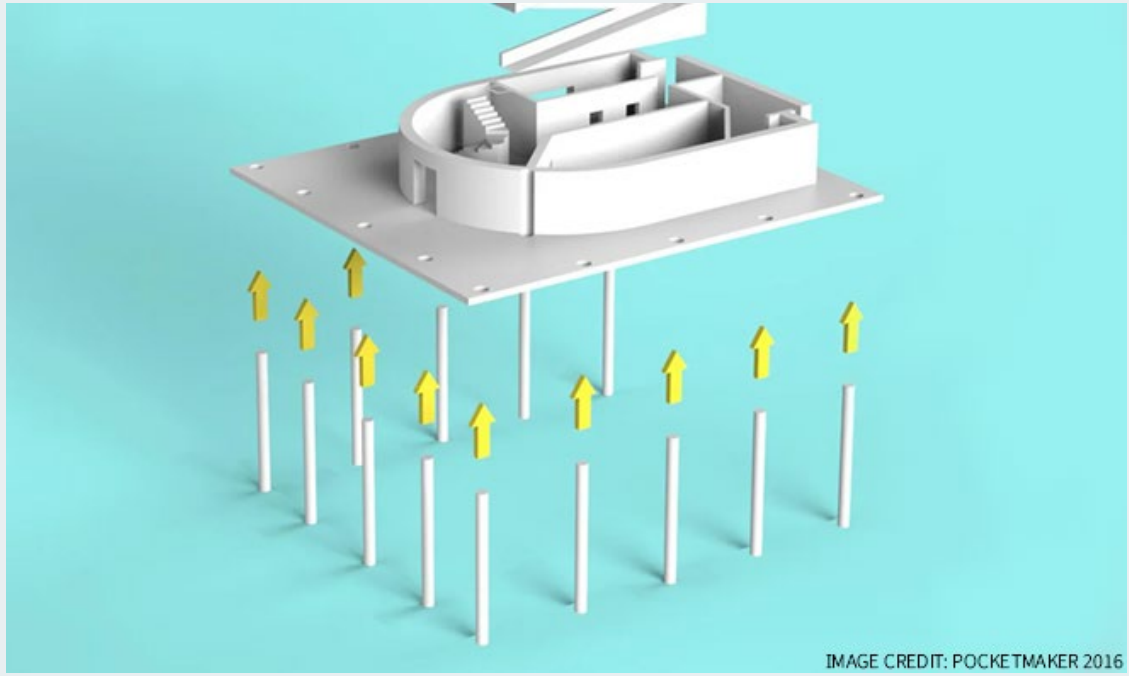
2.1 Abstraction & Level of Detail

- Just like how different scales of an architectural plan warrants more or less information, the level of detail in a 3D print should also only be as high as required for the print scale - is both a conceptual exercise as well as addressing technical limitations of minimal thicknesses and printer resolution as mentioned above.
- You may have to completely remodel certain sections in more/less detail to get the most of the print.
- If you are going down in detail with a mesh, you can also try the [Reduce Mesh] command in Rhino.



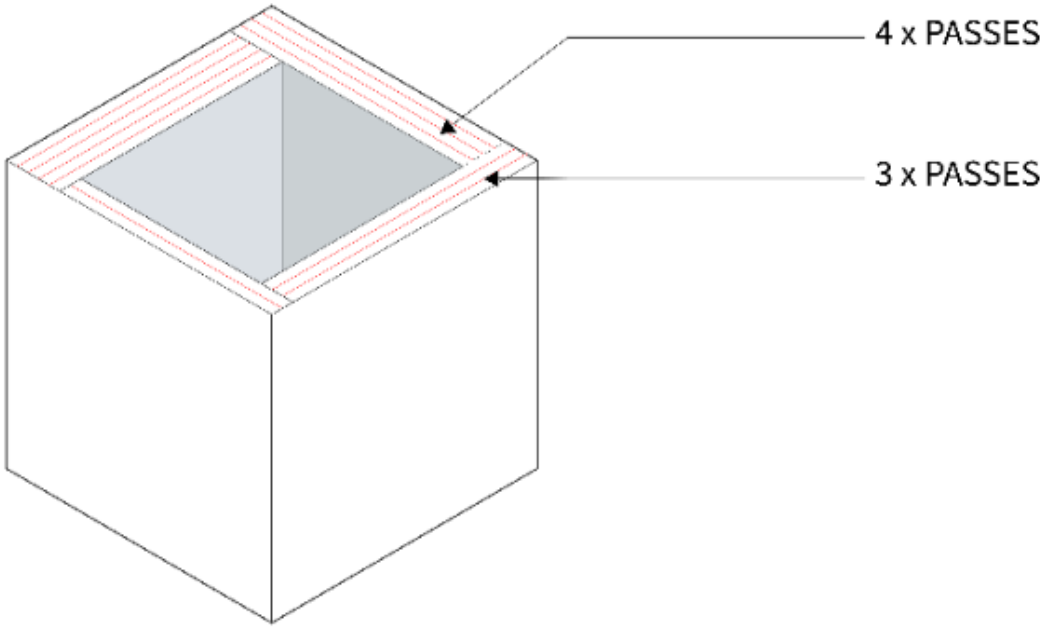
2.2 Substitution

- Substitution can also be used to completely replace 3D printed parts where it is not viable.
- There are many common architectural model-making supplies that can supplement your 3D printing, and actually reduce the hassle of dealing with an unwieldy 3D print.
- This model of the Villa Savoye substitutes out the 3D printed columns with the filament itself - 3D printing columns at this scale would have been extremely fragile at this scale and hard to handle.



2.3 Thickness

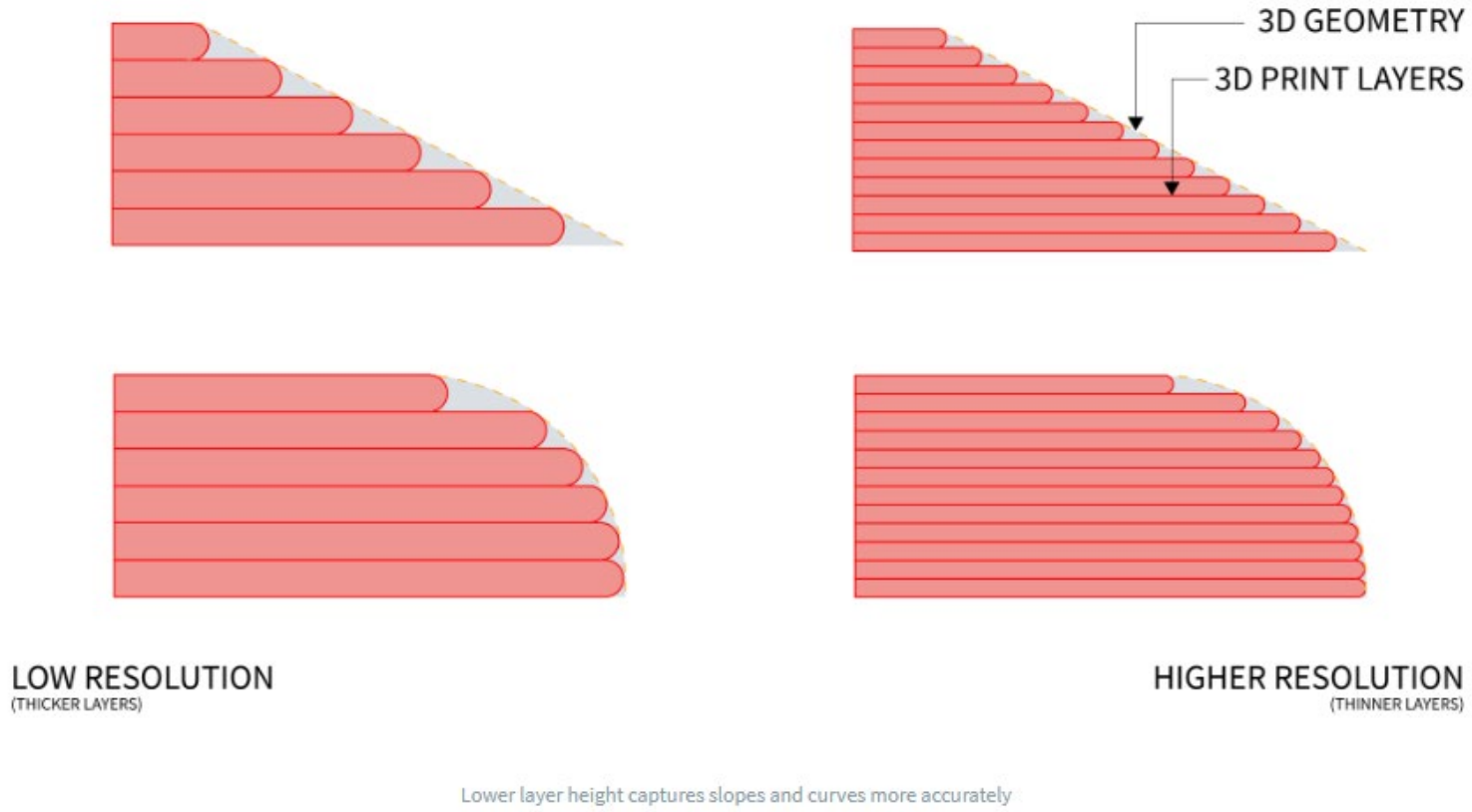
- Thickness is defined by the number of times the extruder will lay filament around the perimeter of your model before it switches to infill.
- Remember! Make walls a multiple of the nozzle size for a smooth slice. E.g. If a printer uses **0.4mm nozzle diameters, leading to 0.2mm, 1.6mm, 2.0mm.**
- **The absolute minimum thickness is 1mm-1.2mm but we recommend a minimum thickness of 2mm.** This will ensure some level of strength, rigidity and good finish for a successful print. This applies to all parts of the geometry, including wall-like and column-like components.
- If working in Rhino, you can look into Mesh Thickening options, or using the sub-geometry selector [Ctrl + Shift + Click] to manually push and pull parts thicker/thinner.



INCREASING SHELLS

2.4 Layer Height Resolution

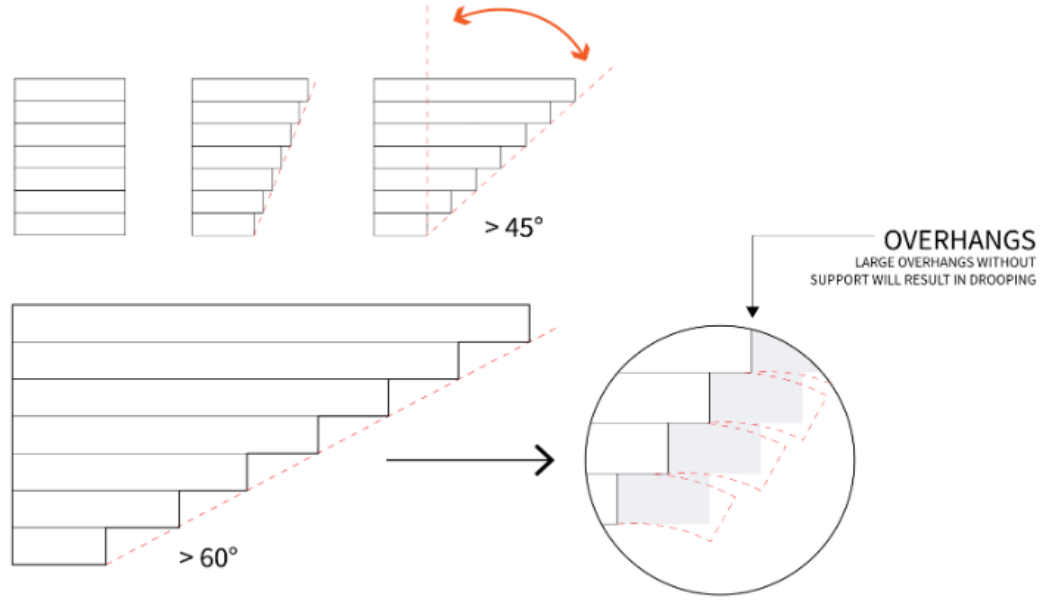
- The layer height is essentially as the name suggests: the height of each layer of plastic deposited during the process. A higher layer height has less resolution but prints faster, while a lower layer height captures more details but is slower.
- 3D printers usually can print down to 0.1mm layer heights. However, bear in mind that in the XY plane, the resolution is limited by the nozzle diameter, which is commonly 0.4mm.



Lower layer height captures slopes and curves more accurately

3. Overhangs

- Overhangs are shapes that extend outwards beyond the previous layer (cantilevers). We know that FDM is a process where each layer is deposited over the previous layer below.
- Therefore, at certain overhanging angles, there may be **insufficient material from the previous layer** for the next layer to print on. This will affect any layers beyond this as it will droop.



Overhang angles at more than 45° will required supports

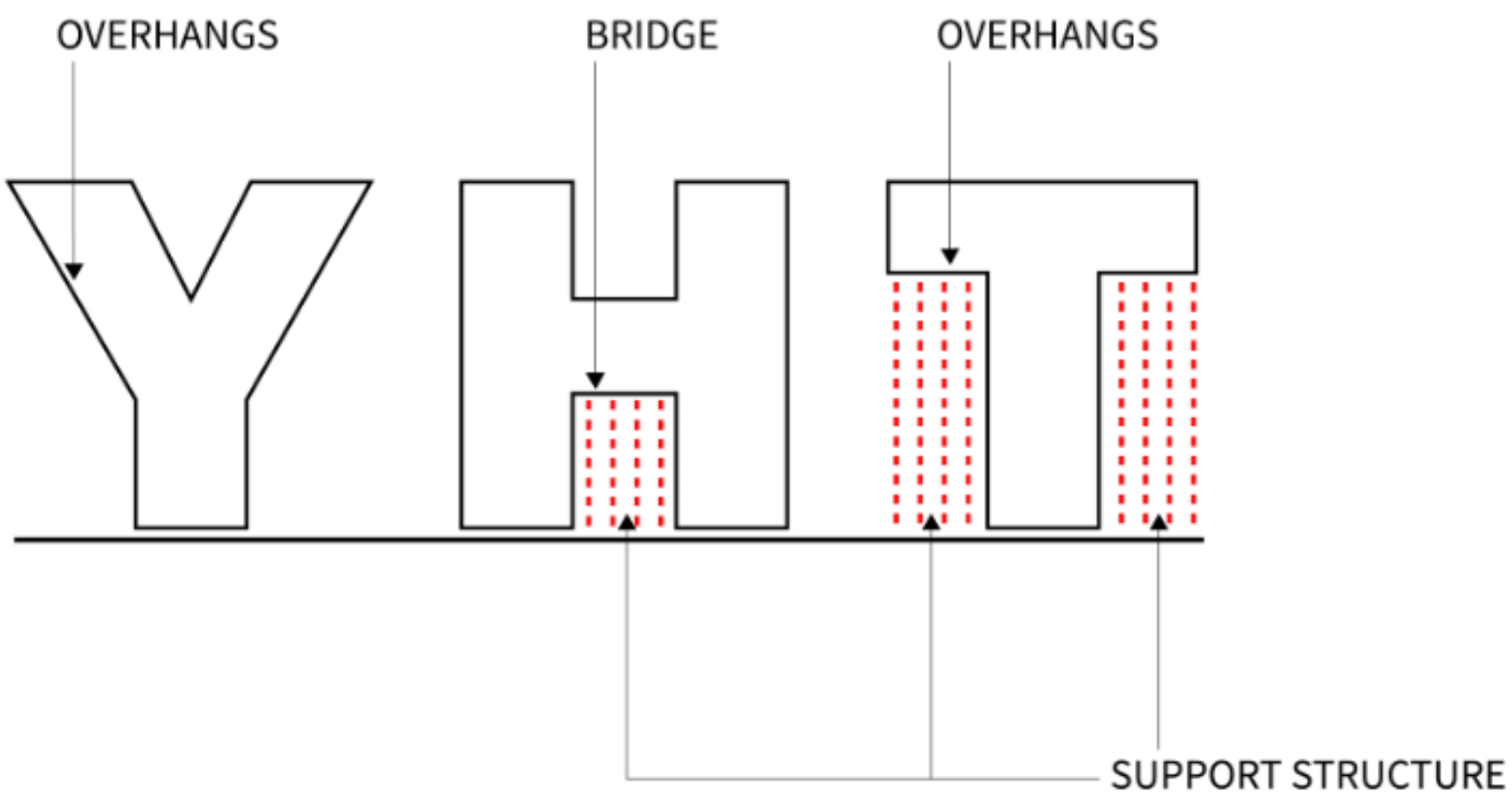
Example with no supports under overhangs



Figure - No Supports under overhangs (3D Hubs)

3.1 Support Material

Supports are sacrificial layers, generated to counter the effects of gravity on filament during printing. Essential part of 3D printing is understanding **support angles, overhangs and bridges.**

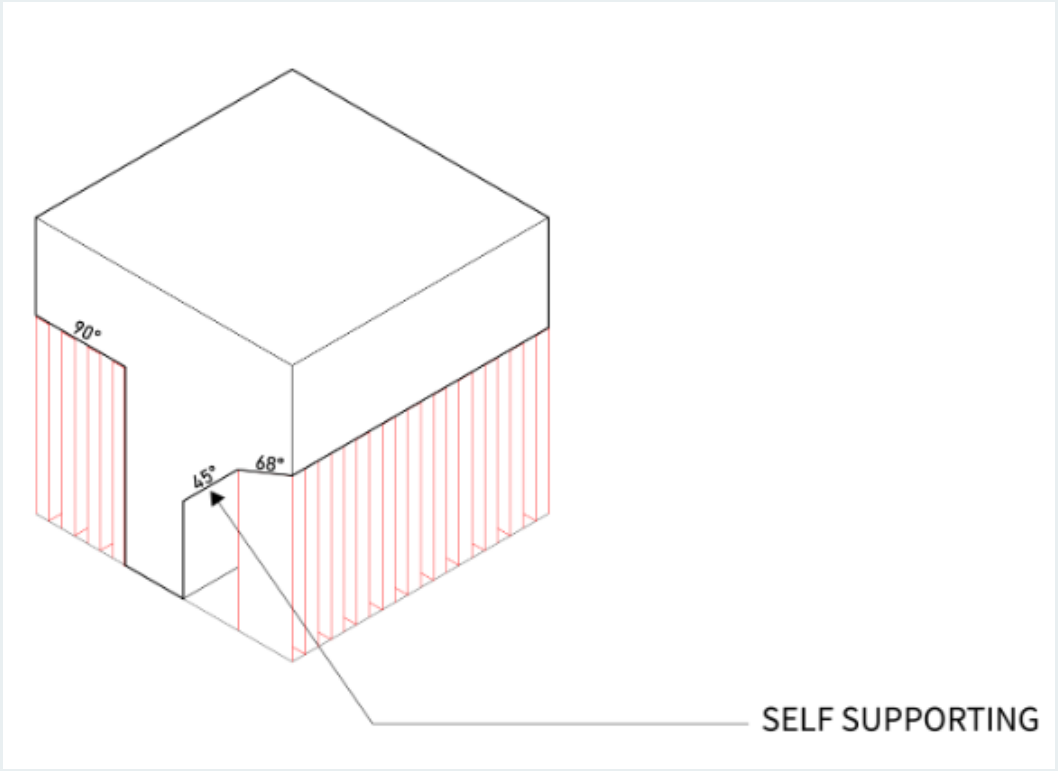


Do I need support?

Angle	Result
< 45°	No intervention required
45° - 60°	Usually no intervention required
60° >	Support Material is required

Self Support or Extra Support?

– In situations where an overhang is necessary and the inclusion of supports will hinder a successful outcome, it is recommended to either incorporate artificial bridges (to be eliminated during post processing), or design the model undersides with a 45° angle.



3.2 Bridges

- Bridging in 3D printing is the horizontal extrusions supported by structures on either sides. Unlike overhangs, you can potentially print bridges without supports; however, this is heavily influence by two main factors: **bridging distance** and **amount of cooling**. Bridges are not perfect; there will inevitably be some **sagging** underneath the bridge as the filament is being extruded over thin air. Some machines can create bridging gaps that span 20mm without supports, with gaps under 2mm having no perceivable sag.

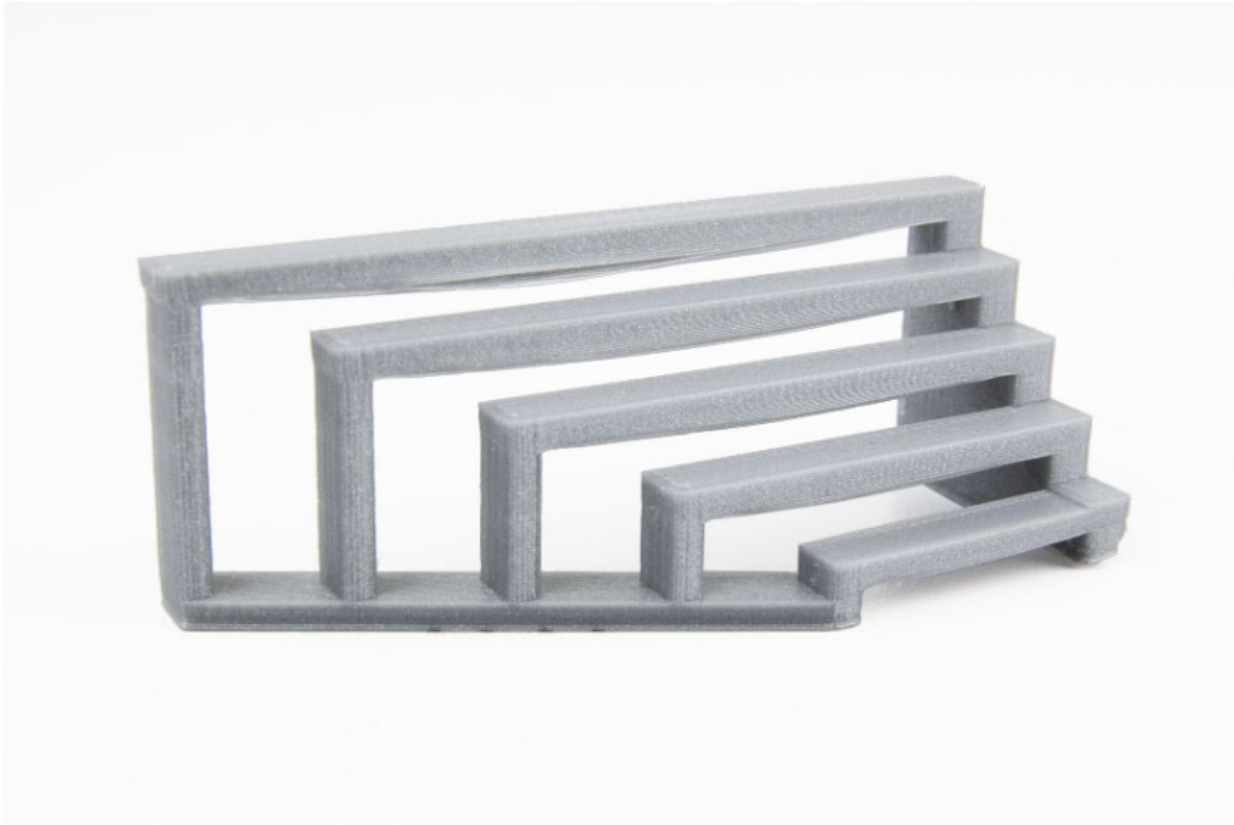
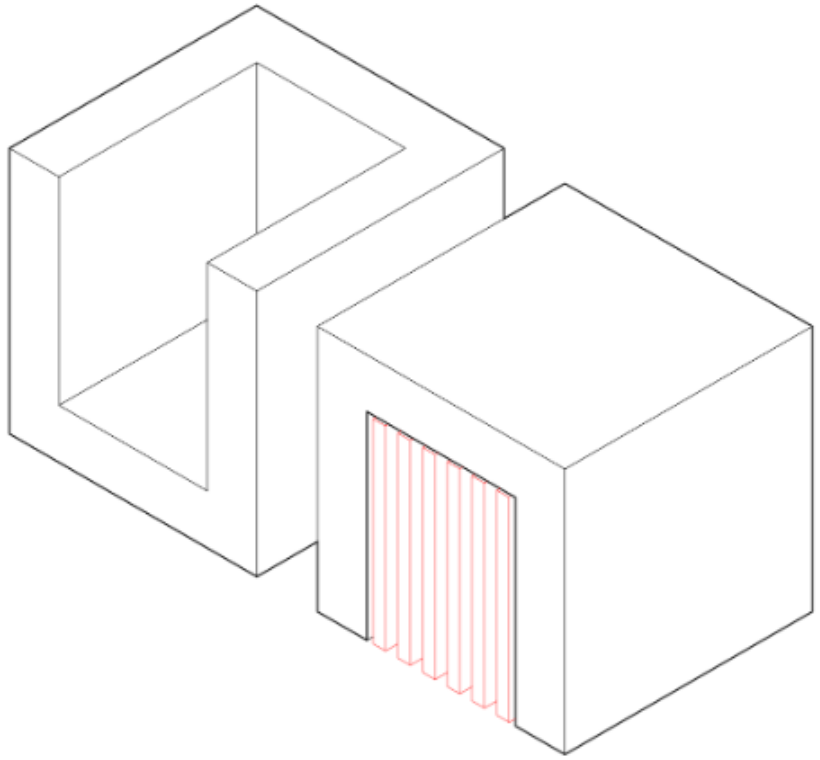


Figure - Bridging (maybe highlight the sagging bits)



ORIENTATION

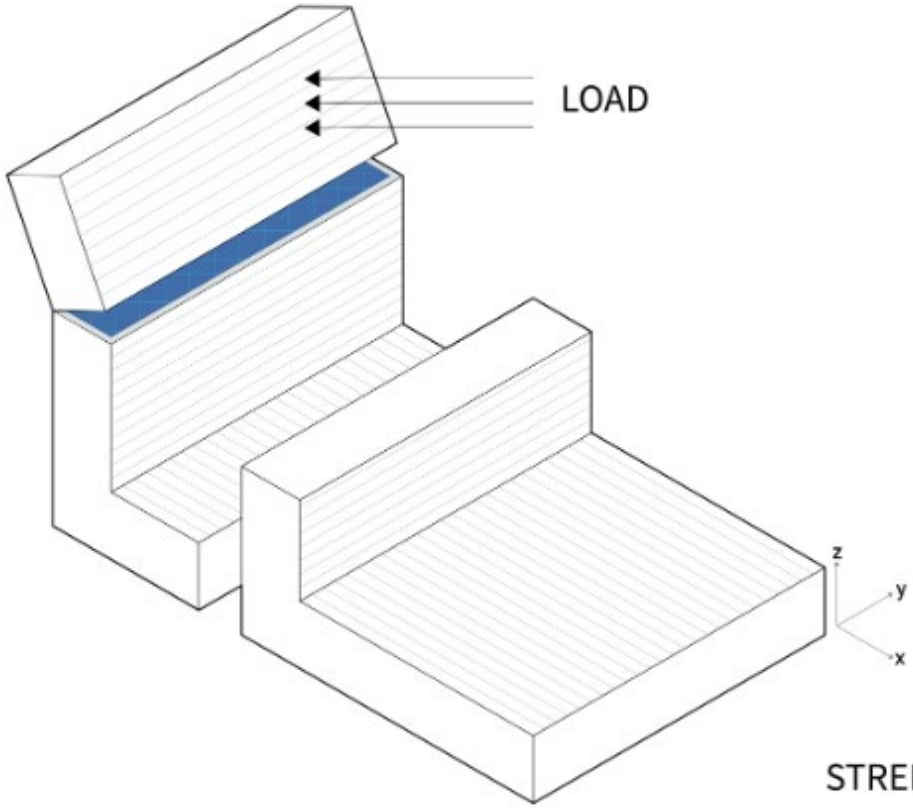
4. Orientation
4.1 Support Material

Ideal orientation minimises support material and achieves the best surface finish. Less material used means a quicker return time and reduces overall cost price.

4. Orientation
4.2 Model Strength

If model strength is key to your application, orienting your model in the X-Y direction will improve stability. The weakest point of a 3D print is where the layers are deposited and adhere to each other.

Orientation may be manipulated as a final step process, where the Print Preview selection could assist in comparing results.

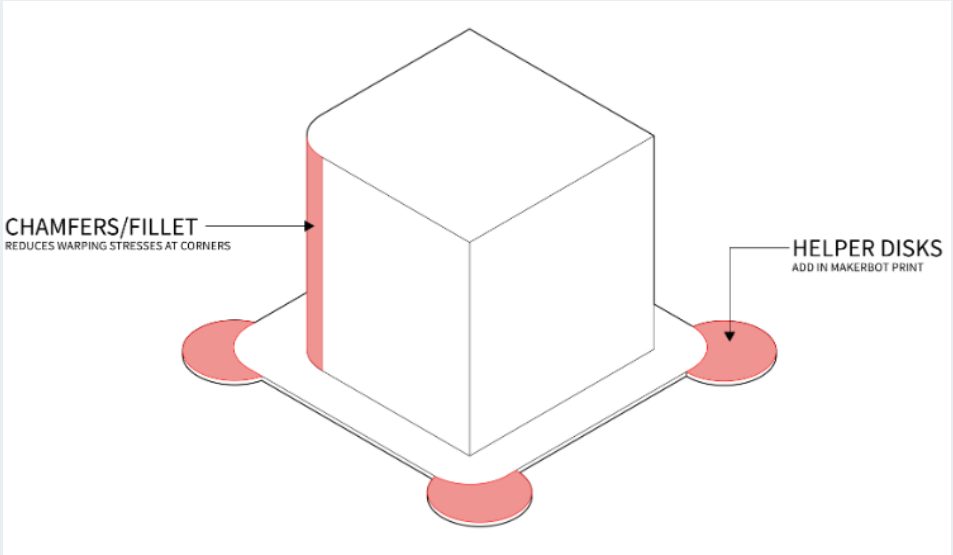


STRENGTH IN ORIENTATION

5. Warpage

As the material expands and shrinks whilst it is being printed, small deformations can happen - if a model requires precision, then these aspects should be considered:

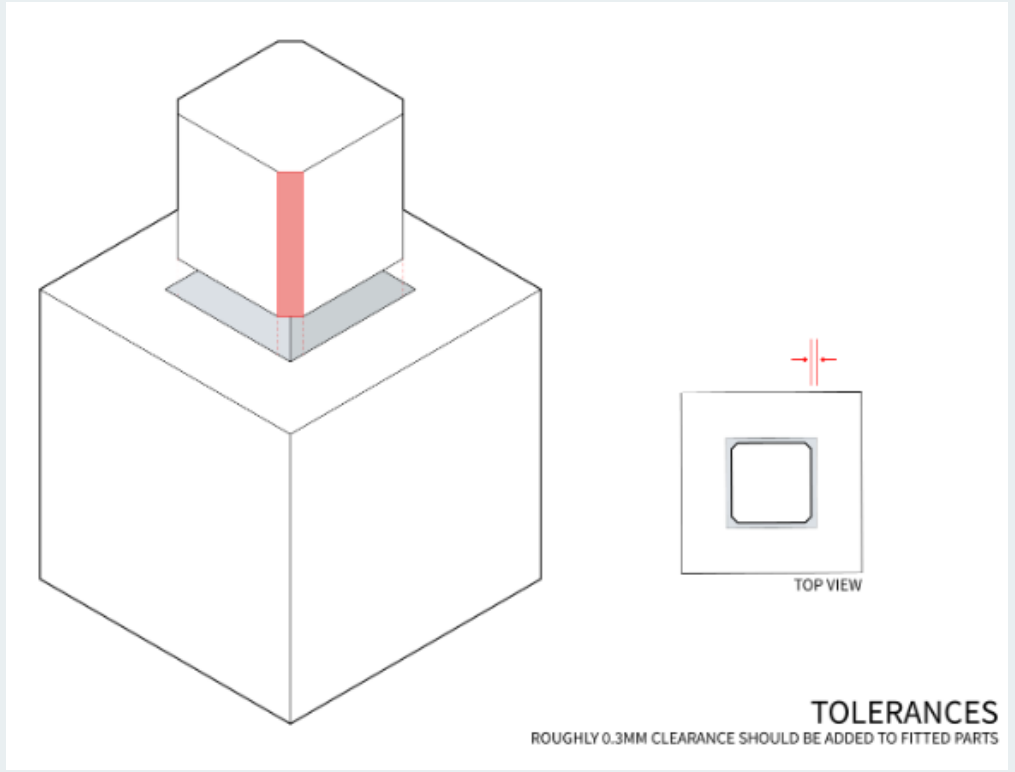
a. Large Surface Area Prints: As your prints increase in size, a key thing to look out for is large surface area prints that may cause **warping**.



5. Warpage

As the material expands and shrinks whilst it is being printed, small deformations can happen - if a model requires precision, then these aspects should be considered:

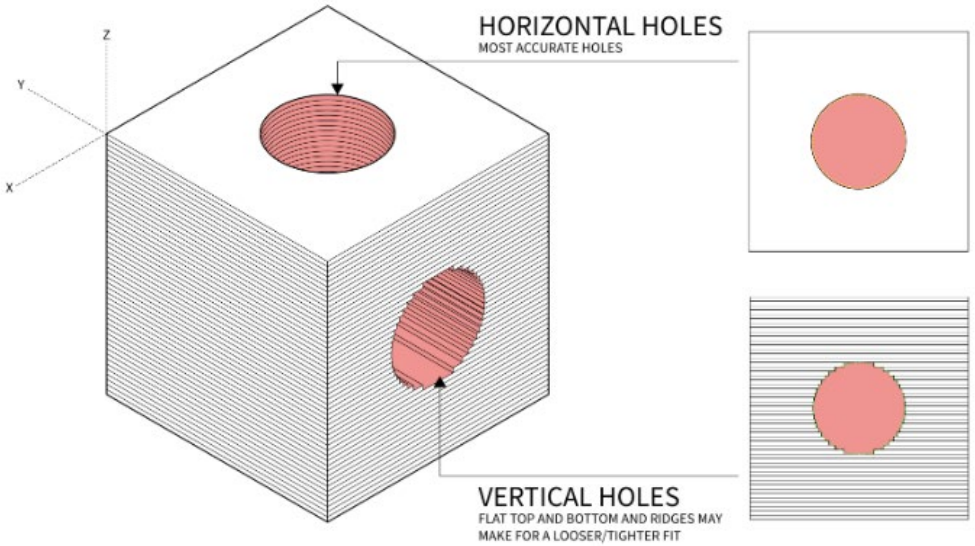
b. Corners: Due to the nature of plastic flow, it is very difficult to achieve true sharp corners in 3D printing. Instead, it is recommended to incorporate fillets and chamfers in your design, particularly with parts that need to fit together. Fillets and chamfers can also help combat the elephant foots effect where the first layer or two that touches your build plate is slightly larger than the rest.
* Rhino Helpful Commands: ChamferEdge, FilletEdge



5. Warpage

As the material expands and shrinks whilst it is being printed, small deformations can happen - if a model requires precision, then these aspects should be considered:

b. Hole Orientation: In general, you want to orient holes to be parallel with the build platform (XY planes). Your holes are most accurate this way. Please note that the fitting of your hole is heavily dependant on tolerances (see Tolerances) and can also be potentially affected by elephant foot effect (see Elephant Foot effect).Sometimes, you will have to print some holes horizontally (XZ and YZ planes). As such, it is important to take note that the finish quality may not be as nice and that tolerances may potentially be looser/tigher (see Tolerances). This is due to contouring nature of 3d printing.



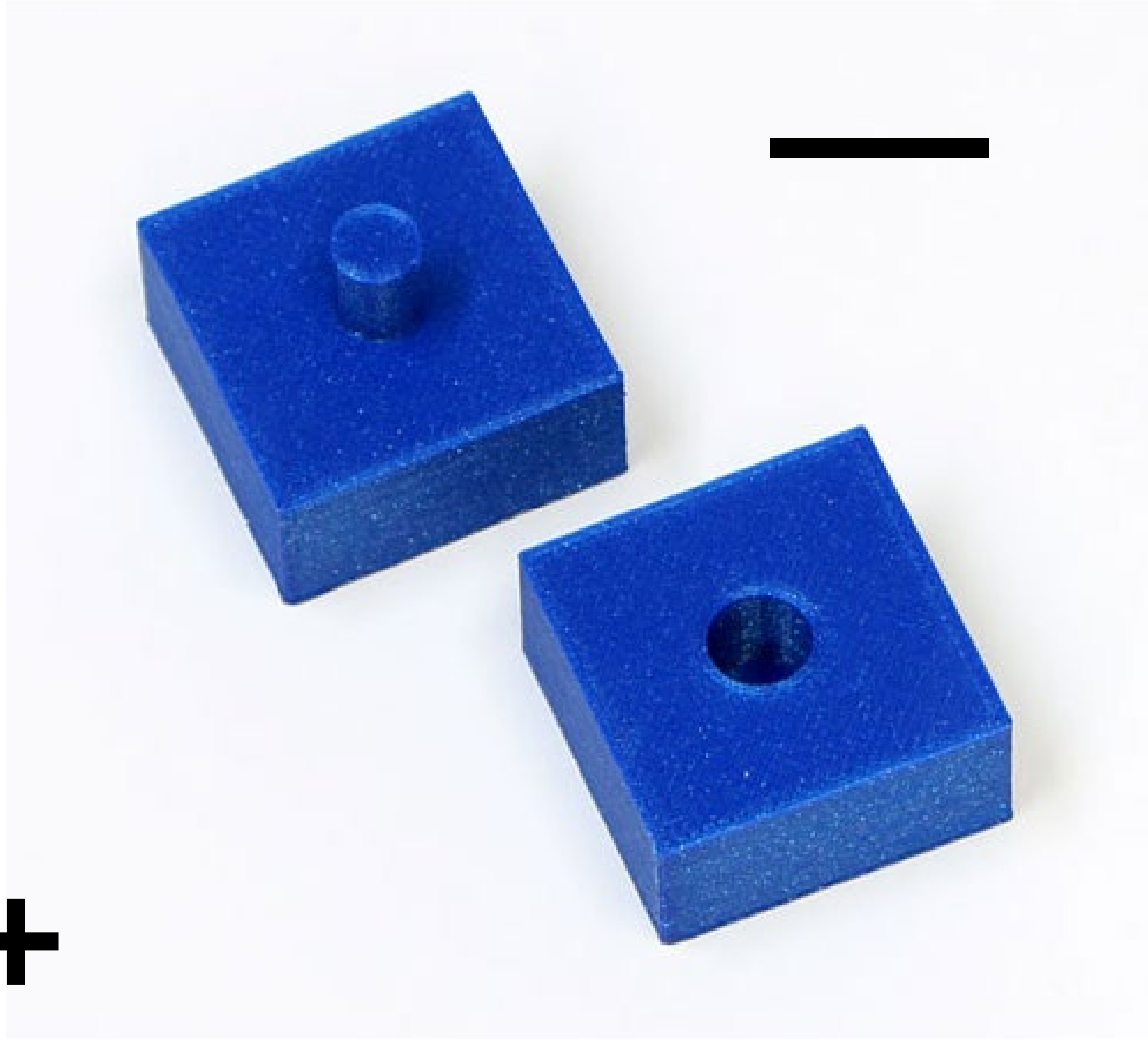
6. Large Models

If your model is too large to fit in the build platform or requires moving joints, you should consider designing alignment and joint elements that allow your split models to come together.



6.1 Alignment Pins

When you split your 3D print into multiple parts, consider using alignment pins/dowels that help connect your parts together. The male part (the pin) should be sized smaller than the female part; a tolerance of 0.5mm is recommended.



Alignment pins.

Reference